

RESEARCH ARTICLE

Data-driven stochastic model for quantifying the interplay between amyloid-beta and calcium levels in Alzheimer's disease

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Abstract

The abnormal aggregation of extracellular amyloid- β ($A\beta$) in senile plaques resulting in calcium (Ca^{+2}) dyshomeostasis is one of the primary symptoms of Alzheimer's disease (AD). Significant research efforts have been devoted in the past to better understand the underlying molecular mechanisms driving $A\beta$ deposition and Ca^{+2} dysregulation. Importantly, synaptic impairments, neuronal loss, and cognitive failure in AD patients are all related to the buildup of intraneuronal $A\beta$ accumulation. Moreover, increasing evidence show a feed-forward loop between $A\beta$ and Ca^{+2} levels, that is, $A\beta$ disrupts neuronal Ca^{+2} levels, which in turn affects the formation of $A\beta$. To better understand this interaction, we report a novel stochastic model where we analyze the positive feedback loop between $A\beta$ and Ca^{+2} using ADNI data. A good therapeutic treatment plan for AD requires precise predictions. Stochastic models offer an appropriate framework for modeling AD since AD studies are observational in nature and involve regular patient visits. The etiology of AD may be described as a multi-state disease process using the approximate Bayesian computation method. So, utilizing ADNI data from 2-year visits for AD patients, we employ this method to investigate the interplay between $A\beta$ and Ca^{+2} levels at various disease development phases. Incorporating the ADNI data in our physics-based Bayesian model, we discovered that a sufficiently large disruption in either $A\beta$ metabolism or intracellular Ca^{+2} homeostasis causes the relative growth rate in both Ca^{+2} and $A\beta$, which corresponds to the development of AD. The imbalance of Ca^{+2} ions causes $A\beta$ disorders by directly or indirectly affecting a variety of cellular and subcellular processes, and the altered homeostasis may worsen the abnormalities of Ca^{+2} ion transportation and deposition. This suggests that altering the Ca^{+2} balance or the balance between $A\beta$ and Ca^{+2} by chelating them may be able to reduce disorders associated with AD and open up new research possibilities for AD therapy.

KEYWORDS

Ca^{+2} dysregulation, Alzheimer's disease, amyloid β peptide, approximate Bayesian computation, Bayesian inference, data-driven models, feedback mechanisms and control, neurodegenerative disorders, neuroscience, stochastic modeling

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1 | INTRODUCTION

Alzheimer's disease (AD) is the most prevalent kind of adult dementia. AD is a medical disorder that gradually kills neurons and produces severe cognitive impairment [1, 2]. Many medication therapies are shown to slow the progression of AD, but there is no permanent cure [3–7]. The clinical and pathological hallmarks of AD include progressive neuronal loss, synaptic degradation, and the formation of amyloid plaques and neurofibrillary tangles in particular regions of the brain [8, 9]. According to some findings, AD may be a systemic disease since it affects not only neurons but also peripheral cells such as fibroblasts, lymphocytes, and platelets in AD patients [10–12].

While the precise etiology of AD remains uncertain, several prominent hypotheses, such as the cholinergic, amyloid cascade, and tau hypotheses, have been proposed to elucidate the development and advancement of AD [13]. The amyloid cascade theory appears to be the most likely, as there are numerous plaques composed of amyloid ($A\beta$) peptide in the AD patient's brain [14, 15]. According to the amyloid cascade theory, $A\beta$ oligomers or amyloid fibrils are formed by the aggregation of $A\beta$ oligomers or amyloid fibrils, which are key components of $A\beta$ peptide and impair the function of neuronal cells [8]. Many mathematical models have been presented to explain the development of $A\beta$ monomer synthesis or aggregation (see [9, 16–18] and the references therein). Another hypothesis that has attracted a lot of attention, proposes that disturbance of calcium (Ca^{+2}) homeostasis is crucial to AD pathogenesis [19–21]. The disruption of Ca^{+2} homeostasis has been extensively explored in order to understand the processes of $A\beta$ -induced neurotoxicity. Intracellular Ca^{+2} operates as a second messenger, regulating neuronal activities such as brain development and differentiation, action potential, and synaptic plasticity [8, 22, 23]. The Ca^{+2} hypothesis of AD proposes that activation of the amyloidogenic pathway affects neuronal Ca^{+2} homeostasis as well as the processes involved in learning and memory. $A\beta$ may alter Ca^{+2} signaling by numerous methods, including boosting Ca^{+2} inflow from the extracellular area and stimulating Ca^{+2} release from intracellular repositories within the brain [24, 25]. Moreover, growing evidence suggests that there is a positive loop between Ca^{+2} and $A\beta$ levels [13, 26, 27]. We know by now, for example, that the persistent high concentration of Ca^{+2} is favorable to the formation of $A\beta$ in rat cortical neurons by imitating $A\gamma$ secretase activity, which is crucial for the breakdown of amyloid precursor protein (APP) [13, 28].

Some of the causes of substantial trial failure include an inadequate understanding of AD etiology and development, as well as an inappropriate trial design. The creation of clinical trial simulations and mathematical modeling of

AD progression are key tools for exploring the reasons why clinical trials fail and refining the clinical trial methodology. The poorly understood nature of AD etiology and development limits the capacity to build solid mechanistic models for reliable disease progression prediction. There are also mathematical models based on inverse problems that have been established to reflect modifications to cognition over time, as measured by errors on various cognitive tests used to assess patients' intellectual capabilities, such as the Modified Mini-Mental State Examination (MMSE) and the AD Assessment Scale [29–36]. The majority of models developed to date in the aforesaid context are stochastic in nature, as evidenced by Ref. [13, 37, 38]. The FDA has authorized following medicines for the clinical treatment of AD, that is, tacrine, rivastigmine, galantamine, and donepezil, which are acetylcholinesterase inhibitors (AChEIs), which increase the concentration and duration of the neurotransmitter acetylcholine's activity (ACh) and another therapeutically utilized medicine is memantine, which is an N-Methyl-d-aspartate (NMDA) receptor antagonist [39]. Additionally, lecanemab and aducanumab are newly approved medicines. In the brains of AD patients, NMDA receptors are overstimulated due to glutamate excess release by neurons, resulting in increased intracellular Ca^{+2} and the death of neuronal cells. Therefore, reducing the concentrations of Ca^{+2} fluxes during the disease state could stop the progression of AD.

Importantly, a prolonged rise in baseline Ca^{+2} could have an essential part in disease development by boosting the production and toxicity of $A\beta$'s in cells with AD-related disorders. These mutually beneficial regulations, in which $A\beta$ promotes a Ca^{+2} increase, which in turn raises the level of $A\beta$, create a positive feedback loop that is expected to create a vicious circle leading to disease development [8, 13, 40]. Inspired by this fact and using clinical data such as study data from the ADNI database (adni.loni.usc.edu), we developed and tested a simple novel stochastic model to predict the interplay between the $A\beta$ and Ca^{+2} concentrations on AD progression in a clinical trial. Also, we have selected the AD patients to be monitored at frequent visits, that is, between 0 and 2 year visits. We analyzed the data for $A\beta$ concentration and fitted it to the developed stochastic model using the approximate Bayesian computation (ABC) technique. ABC is a data-driven strategy that utilizes a number of low-cost numerical simulations. ABC evaluates unknown physical or model parameters and associated uncertainties using reference data from real-world experiments or higher-fidelity numerical simulations [41]. We found that during the disease state in the patient's brain, there is a tremendous increase in $A\beta$ oligomers, which enhance the influxes of intracellular Ca^{+2} . In return, Ca^{+2} encourages the production of these hazardous $A\beta$ oligomers, and this fact reinforces the

positive feedback between Ca^{+2} and $A\beta$. We demonstrate the correlation between the simulations of our model using the ADNI data and the discovery that a variety of dysregulations in Ca^{+2} and $A\beta$ may cause disease, as well as the unpredictability of A + beta in susceptible patients that can cause a change from the “healthy” to the “pathological” state [8, 16, 13, 40, 42, 43, 44, 45, 46, 47]. This vulnerability may explain the high prevalence of sporadic AD found in the elderly population.

In summary, this study presents a novel stochastic model to analyze the interplay between $A\beta$ and Ca^{+2} levels in AD, leveraging ADNI data. Our methodological approach incorporates physics-based Bayesian modeling and ABC to address the complex dynamics of $A\beta$ and Ca^{+2} in AD progression. Through the application of our model, we gained insights into the feedback loop between $A\beta$ and Ca^{+2} , shedding light on potential therapeutic targets for AD treatment.

The rest of the paper is organized as follows: In Section 2, we describe the modeling approach based on the (i) deterministic and stochastic models of the interplay between $A\beta$ and Ca^{+2} , and (ii) the ABC technique. In Section 3, we set up the experimental data and model the participant dynamics. We present our results and computational simulations based on the developed stochastic models in Section 5. The computational results have been obtained with an in-house developed MATLAB code, and all data analysis has been carried out in Python. Finally, we discuss our results, conclude our findings, and outline future directions in Sections 6 and 7.

2 | METHODOLOGY

This section highlights the deterministic and stochastic modeling approaches for AD incorporating the interplay between $A\beta$ and Ca^{+2} within the Bayesian setting. This is achieved by defining the stochastic model of $A\beta$ and Ca^{+2} by adding the stochastic noises. Additionally, we simulate the trajectories of the stochastic model of $A\beta$ and Ca^{+2} using the ADNI data by incorporating the ABC technique presented in Section 2.2. The aim is to fit the ADNI data of $A\beta$ concentrations into the stochastic model of $A\beta$ and see the impact on Ca^{+2} concentrations.

2.1 | The modeling approach

This section will provide the mathematical model for AD that takes into account the inclusion of Ca^{+2} and $A\beta$ [40]. AD is associated with $A\beta$ generated through the fragmentation of the amyloid precursor protein (APP), which is partially lodged in the plasma membrane [40]. APP

is cleaved by either an α - or a β -secretase. In the amyloidogenic pathway, cleavage of APP by the β -secretase generates sAPP β and CTF β . The latter is in turn cleaved by a γ -secretase to form $A\beta$. A rise in cytosolic Ca^{+2} enhances the production and release of $A\beta$ which leads to stimulation of γ -secretase activity in cortical neurons [40]. In resting neurons, the free cytosolic Ca^{+2} level is maintained around 50 – 100 nM, while it is increased up to 1 μ M upon electrical or receptor-mediated stimulation. As described in Ref. [8], Ca^{+2} influx is enhanced by VGCCs or ligand-gated ion channels such as glutamate and acetylcholine receptors. However, the endoplasmic reticulum (ER) is the principal intracellular Ca^{+2} storage, from which Ca^{+2} can be released via the inositol 1,4,5-trisphosphate receptor (IP_3R) or the ryanodine receptor [40]. Decrease of cytosolic Ca^{+2} occurs through Ca^{+2} -ATPases, the Na^+/Ca^{2+} exchanger, or the mitochondrial uniporter [40]. $A\beta$'s perturb the balance between Ca^{+2} entry into and extrusion out of the cytoplasm. In healthy neurons, this process equilibrates, leading to a basal Ca^{+2} level in the range of 50–100 nM [47]. Using transgenic mouse models for AD together with Ca^{+2} imaging, Kuchibhotla et al. [45] have shown that this resting concentration is higher in neurites located close to amyloid deposits, while another study reports that the basal level of Ca^{+2} in the cortical neurons of such animals is around 250 nM, that is, twice that found in controls [48]. Ca^{+2} channels are also deregulated in brain cells and the formation by $A\beta$ oligomers of pores in the plasma membrane enhances the influx of extracellular Ca^{+2} [28]. This feedback is bolstered by the evidence that Ca^{+2} encourages the development of these hazardous oligomers [40].

We will present the basic model illustrated in Figure 1 based on the experimental observations using ADNI data explained in Section 3. The primary factors of the model consist of the levels of Ca^{+2} within cells and the concentrations of $A\beta$ (disregarding any distinctions between intracellular and extracellular compartments or variations in the length and stages of oligomerization of amyloid compounds) [40]. These concentrations are denoted by $A\beta$ and Ca in the equations, respectively. The progression of the two variables of the model equation is given as follows [40]:

$$\frac{d(A\beta)}{dt} = V_1 + V_a \frac{(Ca)^n}{K_a^n + (Ca)^n} - k_1(A\beta), \quad (1)$$

$$\frac{d(Ca)}{dt} = V_2 + k_\beta(A\beta)^m - k_2(Ca), \quad (2)$$

where Ca represents the concentration of the baseline level of cytoplasmic Ca^{+2} that doesn't have significant dependence on the short-lived Ca^{+2} peaks occurring due to the neuronal electrical activity. V_1 represents the constant rate

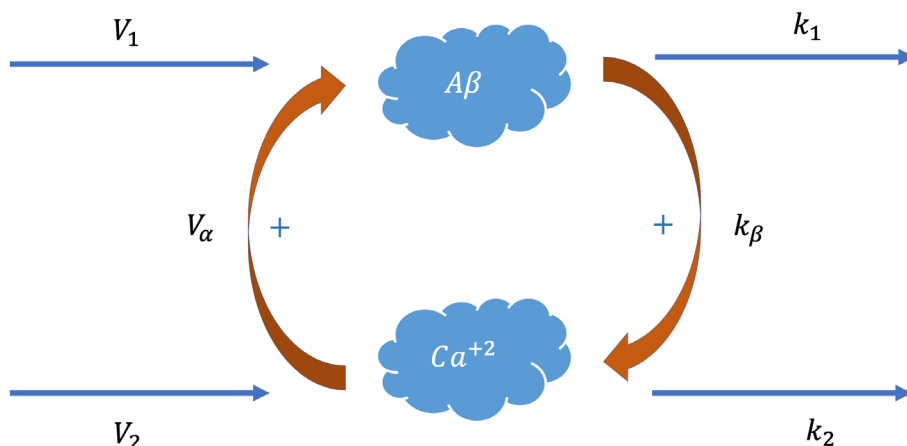


FIGURE 1 Graphic illustration of the mathematical model provided by Equations (1) and (2) that describes the interaction between Ca^{+2} and $A\beta$ during the progression of AD. The positive feedback exerted by Ca^{+2} on the creation of $A\beta$, as well as the fact that $A\beta$ tends to increase intracellular Ca^{+2} , establish a positive loop (motivated by Ref. [40]).

at which $A\beta$ is synthesized and k_1 represents the first-order kinetic rate constant at which $A\beta$ is eliminated. The synthesis of $A\beta$ is activated by Ca^{+2} through a Hill term, characterized by a maximum rate of V_α , a half-saturation constant of K_α , and a Hill coefficient of n . Similarly, V_2 represents the constant rate at which Ca^{+2} enters the cytoplasm, and k_2 represents the first-order kinetic rate at which Ca^{+2} is eliminated [40]. Moreover, Ca^{+2} entry within the cell is induced by $A\beta$ oligomers, potentially owing to the increase in the permeability of the plasma membrane. The cooperativity coefficient m and rate constant k_β represent this process. Notably, the term describing the increase in membrane permeability caused by $A\beta$ oligomers is considered non-saturable to account for the formation of pores by these oligomers [40]. The model Equations (1) and (2) are adopted from Ref. [40]. It is considered that in a healthy neuron, the concentrations of $A\beta$ (~ 5 nM) are lower than those of Ca^{+2} (~ 50 – 100 nM) [49]. In the present section, we formalize this positive loop in a mathematical model and show that it exhibits bistability. Therefore, a stable steady state characterized by low levels of Ca^{+2} and amyloids, which correspond to a healthy situation, coexists with another ‘pathological state’ where the levels of both compounds are high. The onset of the disease corresponds to the switch from the lower steady-state to the higher one induced by a large enough perturbation in either the metabolism of amyloids or the homeostasis of intracellular Ca^{+2} .

It is well known that the majority of models created to date for AD using clinical data in the above context are stochastic in nature [34, 37, 38, 50, 51]. Such models have the important advantage of allowing for variation in model parameters and disease biomarkers when predicting disease progression. Therefore, stochastic noises can be incorporated into the present model of AD that is, Equations (1) and (2), which focuses on the evolution of $A\beta$ and Ca^{+2} . Intrinsic or extrinsic disturbances disrupt the dynamics of $A\beta$ and Ca^{+2} [13]. The intrinsic

noises arise from the random fluctuations of biochemical reaction events such as the stimulation of calcium on γ -secretase activity, the nucleated aggregation process, and the changes of cell membrane integrity induced by $A\beta$ [52], whereas the extrinsic noises originate from the stochastic variations of the microenvironment for $A\beta$ and Ca^{+2} , which include pH , the concentrations of Na^+ , reactive oxygen species, neurons, and peripheral macrophages [40]. Additionally, AD is a neurological disorder that progresses over a long period of time, from a normal state to severe dementia. In contrast, the concentration of Ca^{+2} changes quickly, which is at the timescale of seconds or minutes [13].

On the basis of the above biological backgrounds, we will incorporate the stochastic noises and the explicit time scales into the model Equations (1) and (2) as follows:

$$d(A\beta) = \left(V_1 + V_\alpha \frac{(Ca)^n}{K_\alpha^n + (Ca)^n} - k_1(A\beta) \right) dt + \sigma_1 \sqrt{\epsilon}(A\beta) dB_1(t), \quad (3)$$

$$d(Ca) = (V_2 + k_\beta(A\beta)^m - k_2(Ca)) dt + \sigma_2(Ca) dB_2(t), \quad (4)$$

where $0 < \epsilon \ll 1$ is used to indicate that the change of $A\beta$ concentration is much slower than that of Ca^{+2} concentration and $B_i(t)$, $i = 1, 2$, represent the standard Wiener process defined on a complete probability space $(\Omega, \mathcal{F}, \mathcal{P})$ and $\sigma_i^2 > 0$ for $i = 1, 2$, denote the intensities of white noise, other relevant parameters are adopted from Ref. [13]. The inclusion of a stochastic term in the model serves the purpose of demonstrating that stochastic noises can trigger a sudden transition from a state characterized by lower $A\beta$ concentration to a state characterized by higher $A\beta$ concentration, using ADNI data. These abrupt transitions are significant phenomena in Alzheimer’s disease (AD). Additionally, we examine the effects of stochastic noises on the progression of $A\beta$ and Ca^{+2} , considering that AD models inherently involve stochastic components. The novelty of

the present research is in the development of a stochastic AD model of $A\beta$ and Ca^{+2} using ADNI data for AD patients at 2-year visits (details are given in Section 3). The numerical simulations of the model are anticipated to replicate diverse experimental findings concerning the disease by utilizing the ADNI data. This has the potential to be valuable in the development of therapeutic protocols aimed at decelerating the progression of AD. Since the ADNI data contains only the concentrations of $A\beta$, thus the aim is to fit the $A\beta$ concentration data and incorporate its effect on Ca^{+2} concentrations within the developed stochastic models. This can be done using the ABC technique with details given in Section 2.2.

2.2 | Approximate Bayesian computation (ABC) technique

ABC is a data-driven approach that employs several low-cost numerical simulations. Using reference data from real-world experiments such as ADNI data or higher-fidelity numerical simulations, ABC also estimates unknown physical or model parameters, as well as their uncertainties [53].

Bayesian inference allows for the estimation of instability by evaluating the likelihood of the model parameters supplied by the experimental data [41, 54, 55]. Since our template contains only the $A\beta$ concentrations data, we treat the model Equations (3) and (4) as inverse problems and get data for " $A\beta$ " from the ADNI database for AD patients per 2-year visit. To solve the differential equations (i.e., Equations (3) and (4)) using Bayesian inference, we need to first specify the prior distributions for the unknown parameters (i.e., V_1 , V_α , K_α , n , k_1 , V_2 , k_β , and m), and then use Bayesian methods to update these priors based on the observed data. Let's assume that the priors for all the parameters are independent and normally distributed with a mean of 0 and a variance of 10. This is a fairly non-informative prior that allows for a wide range of possible values for the parameters. Next, we need to define the likelihood functions for the two differential equations. The likelihood function for the first equation (i.e., Equation (3)) is given by:

$$p(\mathbf{A}\beta|\theta_1, t) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(A\beta_i - \widehat{A\beta}_i)^2}{2\sigma^2}\right), \quad (5)$$

where $\mathbf{A}\beta = (A\beta_1, A\beta_2, \dots, A\beta_n)$ is the vector of observed values for $A\beta$ at times $t = (t_1, t_2, \dots, t_n)$, $\widehat{A\beta}_i$ is the predicted value of $\mathbf{A}\beta$ at time t_i based on the current parameter values θ_1 , and σ^2 is the measurement error variance. The predicted values $\widehat{A\beta}_i$ can be obtained by numerically

TABLE 1 List of the parameter values used for deterministic and stochastic models adopted from Ref. [40].

Parameters	Values (units)
V_1	0.0065 (nM/year)
V_α	0.05 (1/year)
K_α	120 nM
n	2
k_1	0.01 (1/year)
V_2	5 (nM/year)
k_β	2×10^{-1} (nM/year)
m	4
k_2	1×10^{-1} (1/year)

solving the differential equation using the current parameter values as given in Table 1. We can use any numerical solver, such as the Runge–Kutta method, to do this. The likelihood function for the second equation (i.e., Equation (4)) is given by:

$$p(\mathbf{Ca}|\theta_2, t) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(Ca_i - \widehat{Ca}_i)^2}{2\sigma^2}\right), \quad (6)$$

where $\mathbf{Ca} = (Ca_1, Ca_2, \dots, Ca_n)$ is the vector of observed values for Ca at times $t = (t_1, t_2, \dots, t_n)$, $\widehat{Ca}_i$ is the predicted value of $\mathbf{Ca}$ at time t_i based on the current parameter values θ_2 , and σ^2 is the measurement error variance. The predicted values $\widehat{Ca}_i$ can also be obtained by numerically solving the differential equation (i.e., Equation (4)) using the current parameter values. To update the priors based on the observed data, we use Bayes' theorem [56]:

$$p(\mathbf{A}\beta, \mathbf{Ca}|\theta, t) \propto p(\mathbf{A}\beta|\theta_1, t)p(\mathbf{Ca}|\theta_2, t)p(\theta) \quad (7)$$

where $\theta = (V_1, V_\alpha, K_\alpha, n, k_1, V_2, k_\beta, m)$ is the vector of unknown parameters. We use a Markov Chain Monte Carlo (MCMC) algorithm, a Hamiltonian Monte Carlo approach implemented in the Python package PyMC3 to sample from the posterior distribution $p(\mathbf{A}\beta, \mathbf{Ca}, t|\theta)$ and obtain estimates of the posterior mean [57, 58]. After inserting the sampled parameters into our model and comparing the resulting simulated uptake values of $\mathbf{A}\beta$ and $\mathbf{Ca}$ with the observed data, we can rate each sample based on its likelihood and use Bayes' theorem to determine the posterior distributions of most likely parameter values for each patient. To fit the ADNI data into the developed stochastic models, we replace the observed data with ADNI data for $A\beta$ concentrations and the time t , as well as the age of the patients given in years. The

details on the ADNI experimental setup data are given as follows.

3 | EXPERIMENTAL SETUP ANALYSIS SUPPORTED BY ADNI DATA

3.1 | ADNI data

The datasets used in the preparation of this article were obtained from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database (adni.loni.usc.edu). The primary goal of the ADNI has been to test whether serial magnetic resonance imaging (MRI), positron emission tomography (PET), other biological markers, and clinical and neuropsychological assessment can be combined to measure the progression of AD. Specifically, we used the ADNI data prepared for the AD modeling challenge and followed the recommendations prescribed in this work to incorporate the interplay between $A\beta$ and Ca^{+2} using the stochastic approach described in Section 2, and fit data for the progression of AD.

3.2 | Modeling the participant's dynamics

We use the qualitative template (available as a CSV file by permission on the ADNI website; adni.loni.usc.edu) for the progression of the AD project dataset, which includes the three ADNI phases: ADNI 1, ADNI GO, and ADNI 2. This dataset contains measurements from brain MRI, PET, CSF, cognitive tests, demographics, and genetic information [50]. From ADNI 1/GO/2, we used the data for 1706 individuals with 6880 visits. The ADNI Conversion Committee made clinical diagnoses of MCI (mild cognitive impairment), NL (normal), EMCI (early mild cognitive impairment), LMCI (late mild cognitive impairment), and AD based on the standards outlined in the ADNI protocol. We designed an analysis based on the data for the individuals with clinical follow-up visits to fall within the 24-month window. Moreover, in the data we use, clinical follow-up visits with improperly arranged dates are discarded for each individual. Also, based on the data statistics, an acceptable time gap of 12 months is estimated for the present study. In the data set, measurements and clinical diagnoses with missing dates and information per visit are set to missing values in order to use the suggested procedure, and participants with fewer than two distinct visits are excluded. Notably, data sets containing missing values and clinical status are denoted as "Missing." For the present study, we have used only the

AD patients' data. We assume that the participants with AD brains for each visit between 2 years have different $A\beta$ concentrations as obtained from the ADNI. Also, we choose the mean of the baseline (bl), 12 and 24 months values of $A\beta$ concentration as the true initial value of $A\beta$, that is, $A\beta_0$ to incorporate the Bayesian inference described in Section 2.2. Therefore, using these initial values first, we will fit the ADNI data to the developed stochastic models, that is, Equations (3) and (4). Then, we will investigate the relationship between $A\beta$ and Ca concentrations by fitting the ADNI data into the developed stochastic models. For the present study, we simply considered only AD patients, given our focus on examining the interplay between $A\beta$ and Ca concentrations during the disease state.

4 | THE SIMULATION ALGORITHM

Our computational approach revolves around employing Monte Carlo techniques to simulate the stochastic process. Leveraging the facilities of the Shared Hierarchical Academic Research Computing Network (SHARCNET: www.sharcnet.ca), our simulation steps for Bayesian inference using ABC is straightforward and can be summarized as follows:

Step 0 (Initialization):

- Initialize time variable: $t = 0$.
- Specify prior distributions for unknown parameters for the n variables $A\beta_1, A\beta_2, \dots, A\beta_n$ and $Ca_1, Ca_2, \dots, Ca_n$.
- Independent and normally distributed with mean 0 and variance 10
- Define measurement error variance: σ^2

Step 1 (Numerical Solution of differential equations):

- Use numerical solver (e.g., Runge-Kutta method) to solve differential equations
- Equation (3) for $A\beta$
- Equation (4) for Ca
- Calculate predicted values
- For $A\beta$: $\hat{A\beta}_i$ at time t_i using current parameter values θ_1
- For Ca : $\hat{Ca}_i$ at time t_i using current parameter values θ_2

Step 2 (Likelihood function calculation):

- Compute likelihood functions for observed data
- Likelihood function for $A\beta$
- Likelihood function for Ca

Step 3 (Bayesian inference):

- Update priors based on observed data using Bayes' theorem

Step 4 (MCMC sampling):

- Use MCMC algorithm (e.g., Hamiltonian Monte Carlo in PyMC3) to sample from posterior distribution
- Obtain estimates of posterior mean for $A\beta$, Ca , and time parameters

Step 5 (Model evaluation and comparison):

- Compare simulated uptake values of $A\beta$ and Ca with observed data
- Rate each sample based on likelihood and determine posterior distributions of most likely parameter values for each patient

Step 6 (Fit ADNI data):

- Replace observed data with ADNI data for $A\beta$ concentrations, time, and patient age
- Incorporate ADNI experimental setup data into stochastic models for further analysis.

In addition to the simulation aspect, our computational procedure encompasses convergence properties and offers a more extensive elaboration on its other attributes. Notably, consistency, or convergence to the true distribution, is not always achieved in this context for all approaches. The empirical distribution linked to the sample serves as the maximum likelihood estimate of the underlying distribution in this more general non-parametric context, where particular parameters or densities are not relevant. Importantly, in our study, a sequence of parameter values converges to the maximum likelihood estimate, giving an output of the algorithm. Leveraging Monte Carlo techniques for simulation, alongside access to SHARCNET supercomputer facilities, our algorithm not only ensures convergence but also exhibits various noteworthy properties worth elaborating on.

5 | RESULTS

In this section, we will present the results obtained from the developed physics-based Bayesian model presented in Section 2. Our model can reproduce typical $A\beta$ and Ca growth dynamics with or without the influence of stochastic noise. In the present study, we are interested in the analysis of such dynamics based on stochastic models of $A\beta$ and Ca (i.e., Equations (3) and (4)), since AD models are predominantly stochastic. Moreover, we have fitted the data for $A\beta$ for AD patients from the ADNI database within 2-year visits in the developed stochastic model. We used the Bayesian inference approach to fitting stochastic differential equations to data. In Bayesian inference, a prior distribution is placed on the parameters, and the posterior distribution of the parameters is

estimated using Bayes' theorem, which takes into account both the likelihood of the data and the prior distribution of the parameters. Once the parameters have been estimated, the equations can be used to simulate the concentrations of $A\beta$ and Ca over time, which can be compared to the experimental data. The latter is a mathematical model of $A\beta$ and Ca that has been formulated within a Bayesian setting to understand the Ca dynamics mediated by $A\beta$ in AD. Note that in Figures 2–4, we produced the results based on the concentrations of $A\beta$ and Ca .

To get an initial insight into the interplay between $A\beta$ and Ca , the simulation of the developed deterministic model (i.e., Equations (1) and (2)) onset of AD describing the positive loop between $A\beta$ and Ca^{+2} is depicted in Figure 5. It can be seen that the model for the onset of AD simulates the shift from a healthy to a diseased state. Here, the stable condition represented by low values of $A\beta$ and Ca^{+2} concentrations is the initial scenario, which corresponds to a healthy state [40]. In contrast, the high values of $A\beta$ and Ca^{+2} concentrations represent the pathological situation [40]. According to the deterministic model (i.e., Equations (1) and (2)), the pathological state may only be attained by long-term changes in Ca^{+2} homeostasis, including those that impact the Ca^{+2} fluxes, as shown in Figure 5. Our findings suggest that when the basal level of Ca^{+2} in the body increases over time, it can decrease the effectiveness of Ca signaling triggered by receptors [59].

Stochastic noises play a crucial role in the metabolism of $A\beta$ and Ca^{+2} [13]. From now on, we will report the results based on the stochastic model, that is, Equations (3) and (4), since the purpose of our study is to fit the ADNI data onto the developed stochastic model. Before fitting the ADNI data we will delineate the results using Equations (3) and (4) based in stochastic simulations as presented in Figures 6 and 2. In Figure 6A, we show that oscillations caused by molecular noise can cause a system to jump from the healthy state to the pathological steady state. Figure 6A illustrates how such a variation might lead to the disease. It's interesting to note that while Ca^{+2} evolves more quickly than $A\beta$, alterations in Ca^{+2} alone are unlikely to cause disease. In other words, the shift to the pathological steady state cannot be reversed by a short-duration of Ca^{+2} rise, which is a characteristic of the model that makes sense given that, without this feature, every action potential would lead to an increase in $A\beta$ as discussed in Ref. [8, 9]. Yet, because of positive feedback and the rapid development of $A\beta$, any noise-induced rise in $A\beta$ will also generate an increase in Ca^{+2} , reinforcing the initial increase in $A\beta$ [40]. Moreover, in Figure 6B, we plot histograms representing stationary density obtained by stochastic simulations

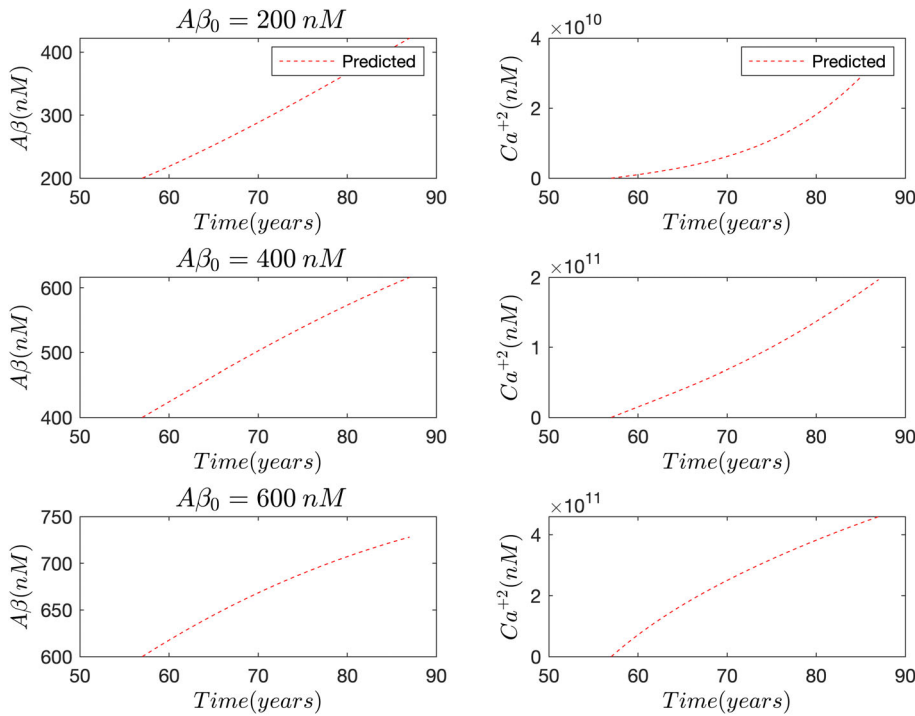


FIGURE 2 The trajectories of the stochastic model created by fitting the simulated data to describe the interplay between Ca^{+2} and $A\beta$ during the start of AD and defined by Equations (3) and (4). The initial values of $A\beta$ chosen here are 200, 400, 600 nM (top to bottom) and for Ca^{+2} , set it to 0. Again, the concentration of $A\beta$ is defined by time scales on a number of years, whereas Ca^{+2} is characterized by seconds to minutes. These values are chosen based on our assumptions to analyze ADNI data, also, $A\beta$ is a promising biomarker that is measured in CSF fluids collected from ADNI participants measured in nM.

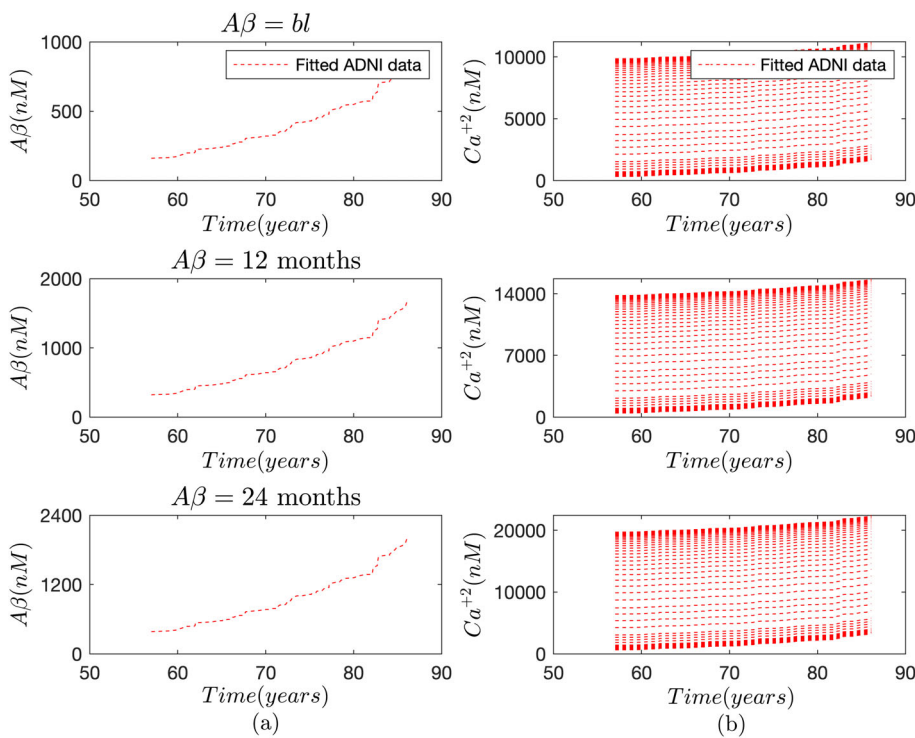


FIGURE 3 The trajectories of the stochastic model by fitting the ADNI data explaining the interaction between Ca^{+2} and $A\beta$ during the beginning of AD and defined by Equations (3) and (4). The initial values of $A\beta$ chosen here are the means of $A\beta$ concentrations at the bl, 12 and 24 month visits (from top to bottom) which were taken as 200, 400, 600 nM. (A) The impact of $A\beta$ (left) on (B) Ca^{+2} (right) is presented. The initial values of Ca^{+2} are set to zero.

of the Equations (3) and (4). We define the number of simulations and the length of each simulation as 1000. The Euler-Maruyama method has been used to simulate the stochastic differential equations for $A\beta$ and Ca^{+2} for a large number of iterations until the system reaches its stationary density [60]. As depicted in Figure 6B, the system of stochastic differential Equations (3) and

(4) follows a normal distribution which shows that the simulated data fit well. We found that, when the noise strength σ_1 and σ_2 increases from zero, the number of extrema of stationary density obtained by stochastic simulations changes. By varying these two parameters, the analysis could investigate how the stationary state properties of the system change with different levels of noise

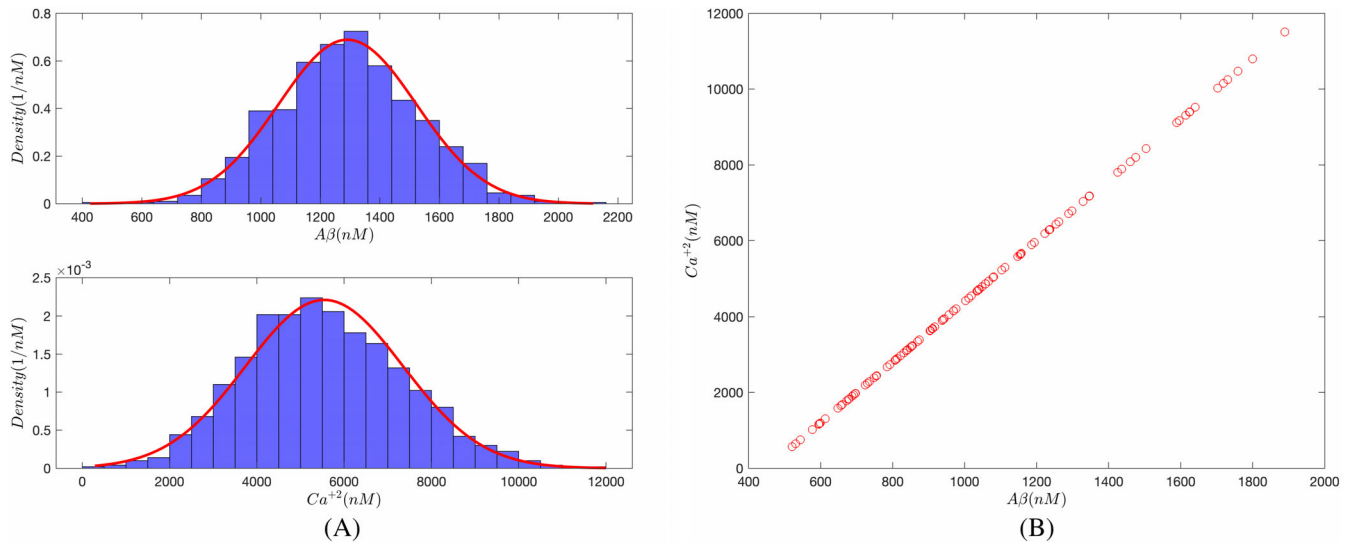
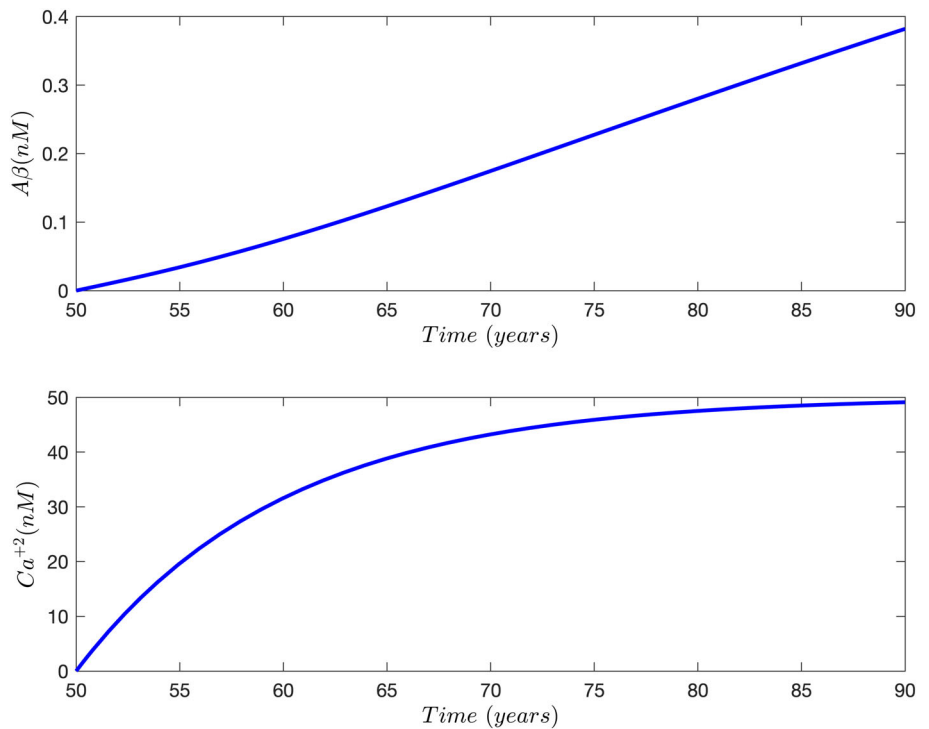


FIGURE 4 (A) The blue histograms represent the stationary density of $A\beta$ (top) and Ca^{+2} (bottom) obtained by stochastic simulations by fitting the ADNI data to the Equations (3) and (4), the red lines stand for stationary density in Equations (3) and (4), the y-axis represents the probability density function (1/nM). (B) The straight line represents the interplay between $A\beta$ and Ca^{+2} concentrations after fitting the ADNI data into Equations (3) and (4). The initial conditions for $A\beta$ and Ca^{+2} chosen here are 400,0 nM, since the patients data has been fitted at 12 month visit (Figure 3). The time scale chosen here is the age of patients ranging from 57 to 87 years based on the ages of patients given on ADNI data.

FIGURE 5 The simulation of the transition in the model for the development of AD from a healthy to a pathological condition, specified by Equations (1) and (2). The transition is caused by a shift in Ca^{+2} homeostasis, which is represented in the model by increasing the rate of Ca^{+2} entry (V_2) from 4 to 5. The initial concentration of both $A\beta$ and Ca has been assumed to be zero, respectively. The parameter values are given in Table 1 and these values were chosen such that the Ca^{+2} concentration, develops quicker than the amyloid $A\beta$ concentration given that the latter is defined by time scales on a number of years, whereas Ca^{+2} is characterized by seconds to minutes (reproduced from Ref. [40]).



strength. For example, the analysis could evaluate the number of extrema of the stationary density obtained by stochastic simulations, as well as other relevant properties such as the mean and variance of the system's state.

Note that in Figures 5 and 6, we produced the results based on the model (Equations (1) and (4)) without fitting

the data. The next goal is to plot the predicted results by fitting the simulated data before adding the ADNI data into the Equations (3) and (4) using Bayesian inference as described in Section 2.2. Therefore, we set up the initial conditions accordingly. At first, the simulated stochastic trajectories were sampled for three different values of $A\beta_0$ for the stochastic model (i.e., Equations (3) and (4)) as

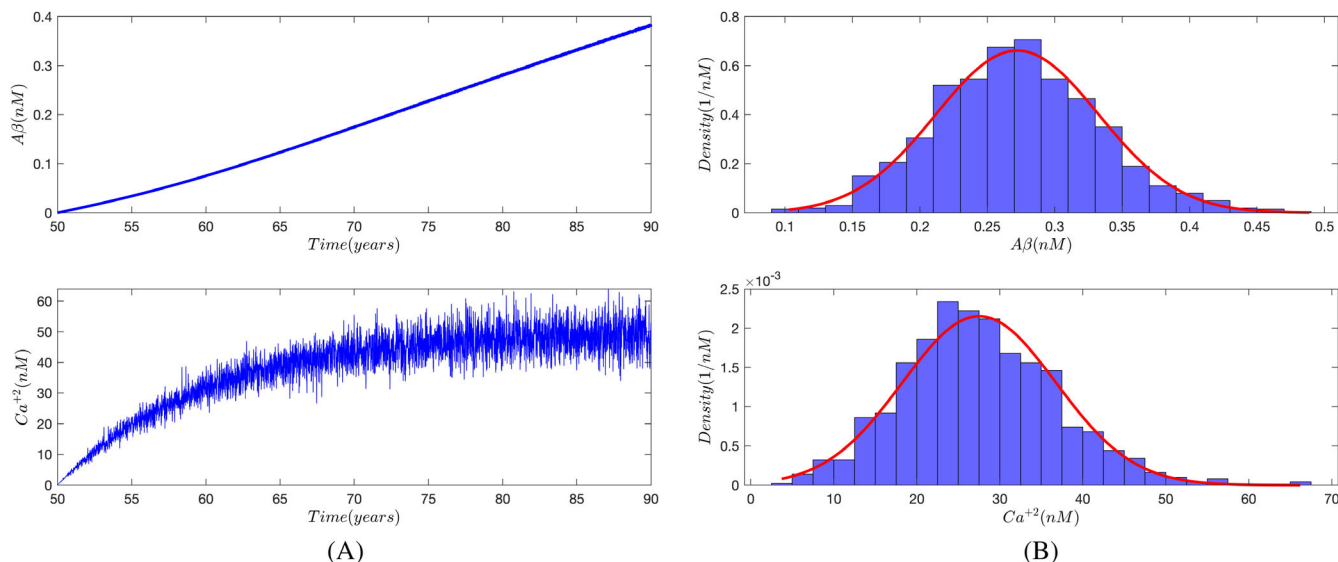


FIGURE 6 (A) The concentration of $A\beta$ and Ca^{+2} in the presence of the Wiener process (i.e., Equations (3) and (4), where $\sigma_1 = 0.2$; $\epsilon = 0.01$; $\sigma_2 = 0.1$). (B) The blue histograms illustrate the stationary density achieved through stochastic simulations, while the red lines depict the stationary density in Equations (3) and (4), the y-axis represents the probability density function ($1/nM$). The relevant parameters are given in Table 1.

presented in Figure 2. These initial values of $A\beta$ are chosen based on the ADNI data (i.e., we choose the mean values of $A\beta$ concentrations at the bl, 12 and 24 month visits as the true initial value of $A\beta$, i.e., $A\beta_0$ for fitting ADNI data) and time is chosen in years that is actually the age of patients. So that the simulated trajectories corresponding to the time periods utilized in the experimental measurements of $A\beta$ completely align. As predicted in Figure 2, we fit the stochastic data using Equations (3) and (4) for $A\beta$ concentration and analyzed its impact on Ca^{+2} concentration. As evident from this figure, the temporal evolution of $A\beta$ concentration is increasing as $A\beta_0$ increases. In return, there is a large jump in the Ca^{+2} concentration and it keeps increasing until the system goes to the disease state. This makes sense that increasing $A\beta$ concentration would increase Ca^{+2} concentration, which leads to the progression of AD. Also, because of the positive feedback and the rapid growth of $A\beta_0$, any noise-induced rise in $A\beta$ will also produce an increase in Ca^{+2} , which will reinforce the increase in $A\beta$ [8, 13, 61].

Importantly, it should be noted that the amount of $A\beta$ is a significant biomarker of AD. The National Institute on Aging (NIA), the Alzheimer's Association, and ADNI have recommended new criteria for diagnosing AD, and one of those suggestions uses $A\beta$ as a biomarker for an early diagnosis of AD [50, 62–64]. Medical studies have demonstrated that AD without symptoms can be found by counting the particles of $A\beta$ in the cerebrospinal fluid or brain [65–67]. With the progress of science and technology, various biomarker tests, including beta-amyloid PET imaging

and cerebrospinal fluid testing, are being utilized in certain settings to assist in the diagnosis by measuring the levels of $A\beta$ [13]. Moreover, work is being done to create straightforward and affordable biomarker tests [50, 68]. Therefore, based on this knowledge, the progression of disease associated with the amount of $A\beta$ used in this study helps predict the likelihood of developing AD because of the significant role that $A\beta$ plays in the diagnosis of AD and Ca^{+2} dysregulation. Our aim is to fit the ADNI data for $A\beta$ concentration using the stochastic model (i.e., Equations (3) and (4)) and see the impact of $A\beta$ on Ca^{+2} concentration for AD patients as per a 2-year visit. In the context of solving coupled differential equations i.e., Equations (3) and (4), ABC involves defining a model that generates the data, choosing a prior distribution for the model parameters, and simulating data from the model. The resulting posterior distribution can be used for inference about the model and the system being studied. Therefore, we have added the ADNI data and solved the inverse problem (i.e., Equation (3)) using the ABC technique as discussed in Section 2.2. Then the resulting data from Equation (3) has been added to Equation (4) to analyze the system for the interplay between Ca^{+2} and $A\beta$ during the onset of AD. The two coupled equations that is, Equations (3) and (4) can be treated as an inverse problem by using Bayesian inference to estimate the values of the unknown parameters that give rise to the observed data. Specifically, we used the available $A\beta$ data on the ADNI database to estimate the disease progression trajectories for Equation (3), and then use these estimated trajectories as input to Equation (4). We can then simulate

the biomarker measurements predicted by Equation (4) and compare them to the observed biomarker measurements. Then we estimated the posterior distribution of the model parameters given the observed data, since in an inverse problem, we are trying to infer the values of the unknown parameters. The corresponding $A\beta$ concentrations with relative contributions of the Ca^{+2} concentrations are evaluated by Equations (3) and (4) and shown in Figure 3. The effects of different values of $A\beta_0$ on $A\beta$ concentrations at bl, 12, and 24 months visits are presented in Figure 3A. The time on the x-axis represents the age of patients per year, assuming that the participants are all 57 years old at baseline as per the ADNI data. It can be seen that there is a rapid growth in the concentration of $A\beta$ for AD patients with three different frequency visits from bl to 24 months. At an initial stage, AD patients have lower $A\beta$ concentrations, but they keep increasing with the passage of time. In return, there is rapid growth in the Ca^{+2} concentrations, which corresponds to $A\beta$ growth during the AD as presented in Figure 3B. This shows that the inclusion of $A\beta$ has altered the steady-state populations of Ca^{+2} due to the interactions between $A\beta$ and Ca^{+2} signaling pathways as depicted in Figure 3. $A\beta$ can disrupt calcium homeostasis by promoting calcium influx and inhibiting calcium efflux, leading to an increase in intracellular calcium levels. This increase in intracellular calcium can alter the steady-state populations of Ca^{+2} by affecting the dynamics of calcium-dependent processes, such as calcium-dependent enzyme activation, gene expression, and synaptic plasticity [20]. Moreover, it is noteworthy that the Ca^{+2} ions stimulate $A\beta$ production, which increases the Ca^{+2} concentrations entering the cytoplasm of neuronal cells, resulting in a positive feedback loop. Interestingly, as the AD patient's age increases, there is a rapid growth in $A\beta$ and Ca^{+2} concentrations since AD progresses. The results and the model predictions obtained in Figure 3 (a-b) align with previous theoretical and experimental studies, which reveal a feedback loop between Ca^{+2} levels and $A\beta$ [13, 40, 43, 44, 45, 46].

Next, to determine whether the ADNI data fit well in the stochastic models (i.e., Equations (3) and (4)) presented in Figure 3, we plotted the histogram as shown in Figure 4A. As depicted in Figure 4A after fitting the ADNI data, we see that both histograms for $A\beta$ (top) and Ca^{+2} (bottom) follow the normal distribution. This demonstrates that the distribution of an ADNI data set follows a normal distribution, with the majority of data points clustering around the mean value and fewer data points in the tails of the distribution. The number of simulations and the length of each simulation (which is in years) are set to 1000. The normal distribution proves that the ADNI data fits well into the Equations (3) and (4). Since the data set follows a normal distribution, it allows

us to make certain statistical inferences and predictions about the data. For example, we can use the mean and standard deviation of the distribution to calculate the probability of observing the experimental data and estimating the unknown parameters (i.e., V_1 , V_α , K_α , n , k_1 , V_2 , k_β , and m) using the ABC technique as described in Section 2.2. In the present study, using Bayesian inference, the likelihood function is used to quantify the probability of observing the ADNI data given a particular parameter value or set of parameter values, as shown in Table 1. Specifically, the likelihood function is a function that takes in the observed data and a set of model parameters and returns the probability of observing the data given those parameter values [69]. Thus, using this approach, the estimated parameters (i.e., V_1 , V_α , K_α , n , k_1 , V_2 , k_β , and m) within the 95% confidence interval are 0.00720, 0.0435, 125, 1.98, 0.02, 4.989, 0.0021, 3.980, 0.099, respectively, by taking the true parameter values as given in Table 1. Importantly, there is a difference in the time scales represented on the x-axis of Figure 6B and Figure 4A. This distinction arises from the specific analysis conducted in each figure. In Figure 6B, we aimed to assess the goodness-of-fit between the simulated data, generated by stochastic simulations of $A\beta$ and Ca^{+2} concentrations as illustrated in Figure 6A. On the other hand, in Figure 4A, we focused on fitting the patient data obtained at the 12-month visit (as shown in Figure 3 for $A\beta$ and Ca^{+2} levels). The objective here was to see the stochastic simulation in the results of simulated data and whether ADNI data fit well or not taking into account the inherent stochasticity of the Wiener process used in the simulations and the time scales of the ADNI data. Therefore, due to the different analyses performed in Figures 4A and 6B, the histograms presented in Figures 6B and 4A have varying time scales.

Finally, the relation between corresponding $A\beta$ concentrations and the relative contributions of the Ca^{+2} concentrations is shown in Figure 4B after fitting the ADNI data. It can be seen that both concentrations are directly proportional to each other. This reveals, that there is positive feedback between Ca^{+2} and $A\beta$. Specifically, as the concentration of Ca^{+2} increases, it leads to an increase in the concentration of $A\beta$, which in turn leads to an increase in the concentration of Ca^{+2} , and so on. This positive feedback loop could be important in understanding the pathological mechanisms underlying AD, which is characterized by the accumulation of $A\beta$ in the brain. This is consistent with an experiment in which Ca^{+2} levels are high during AD and there is positive feedback between Ca^{+2} levels and $A\beta$ concentrations [13, 40, 43–47]. As a result of the bistable tendency [40], the final size of $A\beta$ is determined by the initial levels of $A\beta$ and Ca^{+2} . For example, if the initial concentrations of $A\beta$ and Ca^{+2} are both set

to low values, the system may not reach the threshold concentration needed to trigger the positive feedback loop, and the final size of $A\beta$ would be small. On the other hand, if the initial concentrations of $A\beta$ and Ca^{+2} are both set to high values, the system may quickly reach the threshold concentration, and the positive feedback loop would be initiated, leading to a large final size of $A\beta$. By systematically varying the initial concentrations of $A\beta$ and Ca^{+2} and monitoring the resulting final sizes of $A\beta$, one can observe the dependence of the final size on the initial levels of these concentrations and infer the existence of a bistable tendency [70, 71, 13]. Another strategy in this direction is to assess the dynamics of Ca^{+2} inside neurons as well as any existing alterations to its control during the development of the disease. It has been demonstrated experimentally that there may be a feed-forward loop between $A\beta$ and Ca^{+2} regulation [72]. Mathematically, this has been proven to be a bistable switch, in that when low levels of $A\beta$ and Ca^{+2} (i.e. healthy state) begin to rise due to any form of disruption that leads to increased $A\beta$ or intracellular Ca^{+2} , resulting in certain pathologic effects [40]. This suggests that if the existing levels of $A\beta$ and Ca^{+2} are adjusted to their healthy states, the advancement of AD can be prevented. Because of the positive link between Ca^{+2} and $A\beta$, this might be achieved by reducing Ca^{+2} uptakes. The second technique to delay the development of the disease is to raise the intensity of stochastic noise for Ca^{+2} and reinforce the strength of stochastic noise for Ca^{+2} since these noises tend to lessen the severity of the disease when the stationary distribution is unimodal. They are made feasible by controlling the microenvironment for $A\beta$ and Ca^{+2} . To the best of our knowledge, these are novel results based on the association of $A\beta$ concentration obtained from ADNI data. The interactions between $A\beta$ and Ca^{+2} add a new degree of complexity to important processes related to the beginning and progression of AD and may help to explain why viable therapeutic therapies for the disease have yet to be developed.

The novelty of our present study is that the proposed model utilizes clinical data to measure the relationship between $A\beta$ and Ca^{+2} , specifically ADNI data gathered during per 2-year visits with AD patients. The initial conditions for $A\beta$ and Ca^{+2} are chosen based on our sole assumptions considering the specific data frame since we are only interested in the interplay between $A\beta$ and Ca^{+2} , once the disease started in the individuals. The data was analyzed using the ABC approach and fit our proposed model. The Markov chain Monte Carlo algorithm is used for the model that is fully coupled and for parameterizations. Our research has shown that the presence of $A\beta$ can create a zone with a steady state that is bistable. This means that the system can exist in two stable states, depending on the initial conditions. However, we used the ADNI data

for AD patients only, so we discussed only the disease state and progression of AD. It is believed that the growth of cytosolic Ca^{+2} due to $A\beta$ can lead to the development of AD. Our focus is on the quick development of abnormal Ca^{+2} signals. The concentration of $A\beta$ changes over a much longer period than Ca^{+2} due to accumulation over months, years or even decades. In this case, our research has shown that the presence of $A\beta$ in the model can create a zone with a bistable steady-state population of Ca^{+2} due to the disruption of calcium homeostasis. However, the question of whether the steady-state population of Ca^{+2} in AD patients is bistable requires further investigation and validation using experimental data. Multifidelity modeling is an approach used to efficiently predict the behavior of a complex system by using multiple computational models of varying levels of fidelity or accuracy [73]. In the present study, we used the MCMC approach to combine low and high-fidelity calculations. In this approach, the low-fidelity calculation is used to generate a proposal distribution, which is then refined by the high-fidelity model. Next, the MCMC algorithm iteratively samples from the proposal distribution and accepts or rejects the samples based on their likelihood. This allows the low-fidelity calculations to explore the parameter space efficiently while the high-fidelity calculations are used to refine the results and increase their accuracy. The use of clinical datasets such as ADNI, in conjunction with the computational modeling described herein, facilitates the implementation of multi-fidelity association studies, which are novel and promising tools for evaluating the potential benefits and side effects of therapeutic agents that target known AD pathways. The study has the potential to generate novel ideas and hypotheses for future research, particularly in medication discovery and safety initiatives, which can be validated through additional cohort studies and clinical trials. Moreover, the findings may stimulate further research in the field by highlighting the significance of exploring Ca^{+2} and their potential effects on $A\beta$.

We validated our results using a cross-validation technique which is performed by splitting the $A\beta$ concentrations data at the 24 months visits into five folds and iterating over each fold [74]. We used statistics and a machine learning toolbox, which support cross-validation. We train the model on the training set and return the simulated data and estimated parameters. Additionally, mean squared error (MSE) is computed for each fold, and the mean and standard deviation of MSE values across folds are calculated and displayed. In Figure 7, the scatter plot of true values (i.e., ADNI data fitted values at baseline) vs. predicted values visually demonstrates how well the model's predictions match the actual values. Each point on the plot represents a data point from the testing set. The trend line (red line) overlaid on the scatter plot is a line of best fit,

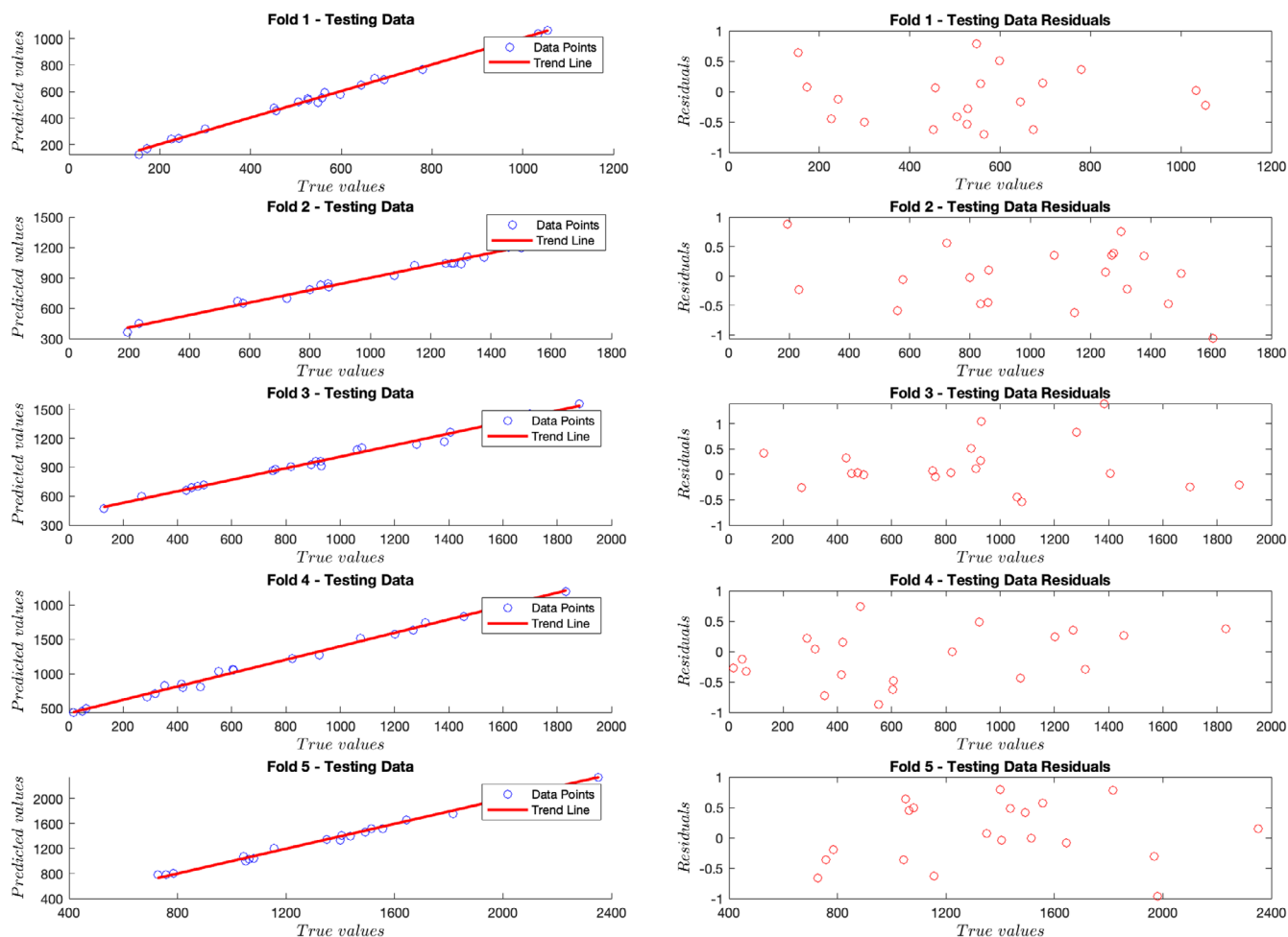


FIGURE 7 The trajectories of the stochastic model by fitting the ADNI data. The true values show the $A\beta(nM)$ concentrations at the 24 month visits. The red line denotes the trend line.

computed using linear regression. It represents the overall relationship between the true values and the predicted values. Therefore, the model predictions are accurate since the points cluster closely around the trend line. Additionally, the residuals plot (residuals vs. true values) shows the vertical distance between each true value and its corresponding predicted value. This plot helps to visualize the model's errors: positive residuals indicate that the model underestimates the true value, while negative residuals indicate that the model overestimates the true value.

The computational results have been obtained with an in-house developed MATLAB code to solve the coupled ordinary differential equations of the developed models (i.e., Equations (1)–(4)) presented in the earlier Section 2. Our data contains 1706 patients, and we divide our job into many processors. We have selected the number of CPUs as the divisors of 1706. For example, for 10,000 iterations, for 1 CPU, the computational time is 5012.1s, and for 2 CPU's it is 140.3s, and so on. To attain the required total time for each model, it often takes several million-time steps. If we

solve the problem using conventional serial programming, this requires a significant amount of computing time. Through the use of open MPI and the C programming language, we can reduce computing time. We divide the sequential tasks involving the time step among available processors for each time iteration and perform them in parallel. All figures have been plotted and shown in Matlab after the data have undergone post-processing. To reduce the time required to acquire results for the parallel computation, we employed the SHARCNET supercomputer facilities (64 cores).

6 | DISCUSSION

In this paper, we extended the stochastic mathematical model of AD by introducing ADNI data for $A\beta$ and analyzed the interplay between $A\beta$ and Ca^{+2} . We investigated the dynamical behaviors of stochastic processes with the model by incorporating slow-fast timescales between $A\beta$

and Ca^{+2} , which revealed the influence of random noises on the advancement of AD. The availability of tools for AD modeling has been relatively restricted thus far, likely because of the immense complexity of the molecular systems involved in its development [40]. Many mathematical models have previously been constructed to explore particular and well-defined features of the disease [9, 16, 17, 18]. To the best of our knowledge, none of the computational models have been proposed yet to explore the synaptic interplay between $A\beta$ and Ca^{+2} using clinical data. Here, we provide a simple stochastic model that qualitatively describes the interactions between intracellular $A\beta$ and Ca^{+2} using ADNI data. It is based on two simple coupled stochastic differential equations with the Wiener process or stochastic noises.

Our goal was to analyze any potential functional effects or interplay between $A\beta$ and Ca^{+2} resulting from the positive loop that exists between the two chemicals in AD patients' brains, despite the fact that the model is obviously oversimplified. Importantly, it is possible for $A\beta$ to bind to NMDA receptors and produce Ca^{+2} dyshomeostasis, which results in oxidative stress, the formation of free radicals, and the death of neurons [75]. Furthermore, $A\beta$ can activate mGluR5 receptors, which elevates postsynaptic Ca^{+2} levels in the cell [8, 49]. Then, APP processing is accelerated by NMDA receptors and mGluR, which create a positive feedback loop that boosts Ca^{+2} influx and free radical generation [76]. Our findings demonstrate that the model effectively accounts for key characteristics of AD, including its irreversibility, the threshold-like transition to severe pathology after a gradual symptom accumulation, the "prion-like" autocatalytic behavior, and the stochastic nature of the disease's emergence in sporadic cases.

Nonetheless, there are several more general characteristics of bistable behavior that may be mentioned. Here, the bistable behaviors mean that the final size of $A\beta$ is dependent on the initial levels of $A\beta$ and Ca^{+2} [13]. Considering that each neuron exists in either one of two distinct steady states, characterized by markedly different levels of $A\beta$ and Ca^{+2} concentrations as well as enzymatic activities, it is anticipated that average measurements of $A\beta$ or Ca^{+2} -related quantities would hold limited significance when it comes to experimental observations on disease characteristics. The model contends that comprehensive identification of the condition of the neurons, or at least of the damaged ones, is required for experimental quantification. Therefore, in the present study, we considered only AD patients, and then in damaged neurons, we analyzed the interplay between $A\beta$ and Ca^{+2} . Our previous study [8] shows that $A\beta$ enhances the dysregulation of Ca^{+2} in AD, so the present studies validate our hypotheses. Furthermore, in terms of the progression of Alzheimer's disease, it is intriguing to note that dysregulations in both $A\beta$ and

Ca^{+2} constitute the causes and effects of the disease in this scenario originating from the positive loop. In this way, by integrating the "amyloid hypothesis," which posits that $A\beta$ is the main causal factor, with the " Ca^{+2} hypothesis," which emphasizes the upregulation of Ca^{+2} signaling as the primary factor, our model effectively reconciles these two frequently debated theories for the onset of AD. Both molecules are intimately connected to one another and equally responsible for the development of AD during the clinical trial, according to the current study.

Our model is limited by the simplification of a single pathway of $A\beta$ changes from many stages (e.g., NL to MCI to AD using ADNI data) to AD. It is well recognized that AD progresses in many ways throughout the years. If large patient samples are researched over an extended period of time, clinical trials are difficult to conduct and expensive. Moreover, modeling and simulating AD dynamics can be done for a small cost, and they are invaluable resources for improving clinical trial designs and raising the probability of accurate treatment efficacy assessments. The results obtained here are consistent with earlier theoretical and experimental investigations [8, 16, 13, 40, 42, 43, 44, 45, 46, 47]. Our goal in this work was to describe the progression of AD, which encompasses not just AD pathology but also biochemical and cognitive alterations brought on by $A\beta$ and Ca^{+2} . The modeling strategy developed here can calculate individual $A\beta$ and Ca^{+2} growth trajectories and markers of latent disease progression at the population level using ADNI data. Individuals are identified along simulated trajectories by utilizing the proposed framework offered by Bayesian inference. This study highlights the rising role of Ca^{+2} ions in the development of AD and focuses on the key components of the interplay between $A\beta$ and Ca^{+2} homeostasis.

7 | CONCLUSIONS

Millions of individuals suffer from the progressive neurological disorder known as Alzheimer's disease. AD patients suffer from gradual, permanent cognitive deterioration. Age represents the most significant risk factor for AD. The hallmark pathological features of AD include the gradual accumulation of cerebral amyloid- β ($A\beta$) peptides, leading to the formation of brain plaques in the extracellular space, as well as the presence of intracellular neurofibrillary tangles composed of misfolded proteins that typically stabilize microtubules within neuronal axons. Additionally, mounting evidence suggests that long-term disruptions in intracellular Ca^{+2} homeostasis may play a crucial role in AD. Specifically, studies have revealed that $A\beta$ contributes to elevated intracellular Ca^{+2} levels, as evidenced by dysregulated calcium dynamics and increased

cytoplasmic Ca^{+2} influx in AD mouse models. Despite extensive research efforts, the precise pathophysiology of AD remains poorly understood, and the underlying molecular alterations associated with the disease have yet to be fully elucidated.

In the present study, we developed a simple, yet effective, stochastic model formalizing a positive feedback loop between $A\beta$ and Ca^{+2} . The novelty of the proposed model is that it incorporates clinical data, such as ADNI data for AD patients per 2-year visit, for quantifying the interplay between $A\beta$ and Ca^{+2} . The data were fitted to the given model using the ABC technique. The goal was to analyze the specific roles of $A\beta$ and Ca^{+2} on synaptic homeostasis and discuss therapeutic protocols to slow down the progression of AD. More specifically, we investigate the underlying mechanisms that lead to neuronal hyperactivity and the role of $A\beta$ growth on the Ca^{+2} dynamics. We demonstrated that in the AD brain, increasing $A\beta$ concentrations could lead to an increase in Ca^{+2} dysregulation, which is harmful and promotes neuronal death. Moreover, there exists a positive feedback loop between the growth of both compounds (i.e., $A\beta$ and Ca^{+2}). Importantly, we validated our results using the cross-validation technique. It is expected that the proposed model will assist in a more precise prediction of the synaptic mechanism during AD and pave the way for the experimental testing of different hypotheses. We provided numerical simulations that agree with the previous findings that a number of dysregulations within the brain can lead to a disease state [8, 16, 42, 43, 44, 45, 46, 40, 13, 47]. Importantly, changing the balance between $A\beta$ and Ca^{+2} concentrations or lowering both concentrations may be able to alleviate AD-related disorders and open up new research avenues for AD treatment. Our findings fill gaps in AD research by explaining how $A\beta$ plaques develop, what happens when Ca^{+2} and $A\beta$ interact, and how they induce selective neuronal death in AD patients. Future research will be able to more precisely evaluate model predictions and forecast the progression of the disease by including more patients, for instance, MCI, EMCI, and NL, in clinical trial data.

AUTHOR CONTRIBUTIONS

HS: Methods and materials, data curation, formal analysis, investigation, writing, and original draft preparation. **SS and RM:** Designed research. **SS:** Methodology, review and editing, and supervision. **RM:** Conceptualization, supervision, and reviews. All authors approved the final submitted version.

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CONFLICT OF INTEREST STATEMENT

The authors declare no competing interests.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from Alzheimer's Disease Neuroimaging Initiative: ADNI. Restrictions apply to the availability of these data, which were used under license for this study. Data are available from <https://adni.loni.usc.edu/data-samples/access-data/> with the permission of Alzheimer's Disease Neuroimaging Initiative: ADNI.

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